Technical Design Note

Power Electronics and Control Systems

CLLC-DCX isolated DC/DC stage

800 V / 800 V, 50 kW, 50 kHz: sizing, simulation model and library block

Isolated DC transformer of the solid state transformer module, with the 35 kHz *variant* 790 V *to* 700 V*.*

Authors:	**Revision:**	**Date:**
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Abstract: The solid state transformer (SST) module of the `solid-state-transformers` repository couples a 800 V battery bus to a single-phase T-type inverter through a galvanically isolated DC/DC stage. This note documents that stage: a bidirectional CLLC resonant converter operated as a DC transformer (DCX), that is at fixed 50 % duty, with the switching frequency a little below the series resonance and the rectifying bridge driven as a synchronous rectifier by the sign of its own current. The sizing of the 800 V / 800 V, 50 kW, 50 kHz stage is derived step by step (operating point, resonant tank and magnetizing inductance, transformer, semiconductors, DC links, loss budget), together with the reasons why the stage is left in open loop. The 35 kHz variant 790 V to 700 V, kept in the repository next to the specification, is documented in the same way and the origin of its numbers is explained. The second part describes the Simscape simulation bench of the repository (power flow modes, switch model, numerical settings and the numerical damping of the fixed-step solver) and the compact library block `cllc_dcx` of `pwr-control/library`, a single Simscape component for the power stage and a single MATLAB Function for modulator, synchronous rectifiers and control, with its ports, measurement and gate vectors, control log and parameter structure.

Revision history

Rev.	Date	Description	Authors
00	October 2026	First issue: sizing of the 50 kHz stage and of the 35 kHz variant, simulation bench, library block.	Davide Bagnara

Project Repository: solid-state-transformers.git

Framework: MATLAB/Simulink/Simscape

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Nomenclature

Symbol	Meaning
u_1, u_2	DC voltages of side 1 (battery bus, 800 V) and of side 2 (isolated DC link of the inverter)
i_{dc1}, i_{dc2}	DC currents of the two sides; i_{dc1} positive into the stage, i_{dc2} positive out of it
P	transferred power, 50 kW nominal
$n = n_1/n_2$	turns ratio of the transformer (11 : 11 for the 50 kHz stage, 9 : 8 for the 35 kHz variant)
f_s, ω_s	switching frequency of the bridges and its angular value
f_r, ω_r	series resonance of one side of the tank, $\omega_r = 1/\sqrt{L_{r1}C_{r1}}$
$f_n = f_s/f_r$	normalized switching frequency, 0.95 in both designs
L_{r1}, C_{r1}	resonant inductance and capacitance of side 1 (leakage plus discrete inductor)
L_{r2}, C_{r2}	resonant inductance and capacitance of side 2, $L_{r2} = L_{r1}/n^2$, $C_{r2} = n^2 C_{r1}$
L_m	magnetizing inductance of the transformer, referred to side 1
$L_n = L_m/L_{r1}$	inductance ratio of the tank
$Z_0 = \sqrt{L_{r1}/C_{r1}}$	characteristic impedance of one side of the tank
$R_{ac} = 8n^2u_2^2/(\pi^2 P)$	first-harmonic (FHA) load resistance seen from side 1
$Q = Z_0/R_{ac}$	quality factor of the tank at the operating point
M	FHA voltage gain of the tank, $M = n\,u_2/u_1$
$X(f_s)$	series reactance of the whole tank at the switching frequency, referred to side 1
φ	phase displacement of bridge 2 after bridge 1 (positive when bridge 2 lags)
i_{r1}, i_{r2}	tank currents of side 1 and side 2
I_{1pk}, I_{1rms}	peak and rms value of the resonant current of side 1
I_m	peak of the magnetizing current
u_{b1}, u_{b2}	output voltages of the two full bridges (square waves of amplitude u_1, u_2)
u_{cr1}, u_{cr2}	voltages across the two resonant capacitors
C_{oss}	output capacitance of one switch, energy related
t_d	dead time of the driving bridge, 300 ns
t_c	step of the Simscape local solver and sample time of the control block

Symbol	Meaning
R_{num}	extra series resistance introduced by the Backward Euler method at the switching frequency
C_{dc1}, L_{dc1}	DC-link capacitor and DC choke of side 1 (the same on side 2 with index 2)
DCX	DC transformer: a resonant converter run at fixed frequency and fixed 50 % duty
SR	synchronous rectifier (the bridge that rectifies, driven by the sign of its own current)
FHA	first-harmonic approximation
ZVS	zero-voltage switching

1. Introduction and scope

The solid state transformer studied in [1] is a chain of modules, each one made of an isolated DC/DC stage and of a single-phase inverter; the DC/DC stages share the battery bus and the inverters are connected in series on the AC side. The repository `solid-state-transformers` [3] keeps, in the folder `theory_analysis_solid_state_transformer`, the simulation bench of one module built around the two stages documented here and in [2]: the CLLC resonant converter used as a DC transformer (this note) and the single-phase T-type inverter with its grid-tied control.

Figure 1 places the DCX in the module. Side 1 of the DCX is the 800 V battery bus of the module; side 2 is the galvanically isolated 800 V DC link of the T-type inverter. The DCX has three tasks and no other: it provides the galvanic isolation that lets the inverters be stacked in series, it carries 50 kW in both directions, and it keeps the ratio u_2/u_1 fixed at $1/n$. It does not regulate: the voltage of the inverter DC link follows the battery, and the current drawn from it is set by the inverter. This is what the term DCX means [5, 8], and it is the reason why the stage can run at the optimum of the resonant tank at any load, with zero-voltage switching of the driving bridge and almost zero-current switching of the rectifying one.

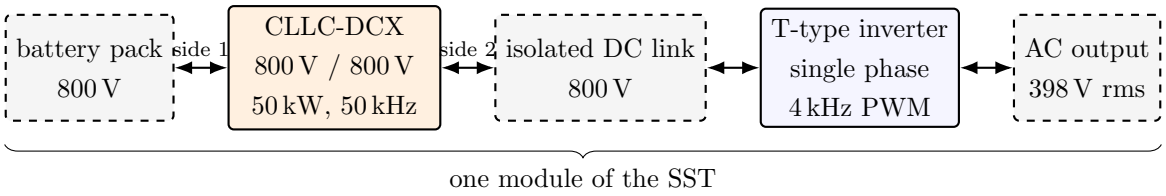


Figure 1: The DCX in the SST module: side 1 on the battery bus, side 2 on the isolated DC link of the single-phase inverter. The DCX fixes the ratio of the two voltages; the inverter sets the power. The bench of [3] puts two modules in series on the AC side.

The note has two parts. Sections 2 and 3 are the sizing: the 800 V / 800 V, 50 kW, 50 kHz stage that is the specification of the module, and the 35 kHz variant 790 V to 700 V that the repository keeps next to it. Sections 4 and 5 are the models: the Simscape bench of the repository, built with the discrete blocks of the library, and the compact block `cllc_dcx` of `pwr-control/library` [4], which packs the whole stage into two nonvirtual blocks for the SST bench. The numbers of this note are those of the parameter files of the two repositories (`private/cllc_dcx_param.m` of each bench folder, `cllc_dcx_param_example.m` and `cllc_dcx_param_example_35kHz.m` of the library); the analytic figures are produced by the script `py/cllc_dcx_figures.py` of this folder from the same numbers. At the time of writing no simulation run of the DCX is saved in the repository: Section 4.5 lists what the first runs have to compare with the sizing.

2. Sizing of the 800 V / 800 V, 50 kW, 50 kHz stage

2.1. Operating point

Table 1 is the operating point. Both DC links are at 800 V, so the transformer is 1 : 1 and the same SiC module is used on both bridges; the base current is $i_{\mathrm{dc1}} = P/u_1 = 62.5\,\mathrm{A}$ on both sides. The switching frequency is 50 kHz, the highest value at which the transformer still fits in one layer of litz wire per winding on a stack of standard ferrite U-cores (Section 2.5) and at which the 300 ns dead time of the SiC modules is still a small fraction of the period (5.4° of the fundamental).

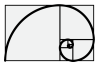


Table 1: Operating point of the DCX.

Quantity	**Value**	**Note**
DC voltage, side 1 (u_1)	800 V	battery bus of the module
DC voltage, side 2 (u_2)	800 V	isolated DC link of the T-type inverter, $u_2 = u_1/n$
Turns ratio	11 : 11	one layer per winding, Section 2.5
Nominal power	50 kW	both directions
DC current, both sides	62.5 A	base of the pu system of the model
Switching frequency f_s	50 kHz	both bridges, 50 % duty
Series resonance f_r	52.6 kHz	$f_n = f_s/f_r = 0.95$
Dead time t_d	300 ns	driving bridge, ZVS with the magnetizing current
Semiconductors	Wolfspeed CAB006M12GM3	1200 V SiC half-bridge modules, both bridges
Control	open loop (DCX)	bridge 2 synchronous rectifier on i_{r2}

2.2. Principle of operation and first-harmonic gain

The CLLC converter [5] is the bidirectional form of the LLC converter [7, 8]: two full bridges and a transformer whose leakage, completed by a discrete inductor, resonates with a series capacitor on each side (Figure 5). The two tanks are symmetrical when referred to the same side, $L_{r2} = L_{r1}/n^2$ and $C_{r2} = n^2 C_{r1}$, so the converter has the same resonance and the same gain in both directions of the power flow.

With the first-harmonic approximation (FHA) [6] the load is the resistance $R_{ac} = 8n^2u_2^2/(\pi^2 P)$ and the gain of the tank, referred to the primary, is

$$M(f_n) = \frac{n\,u_2}{u_1} = \frac{1}{\sqrt{\left[1 + \frac{1}{L_n}\left(1 - \frac{1}{f_n^2}\right)\right]^2 + Q^2\left(f_n - \frac{1}{f_n}\right)^2}}, \qquad Q = \frac{Z_0}{R_{ac}}, \quad L_n = \frac{L_m}{L_{r1}}, \tag{2.1}$$

which is the gain of the LLC tank with the load resistance in series with the second resonant branch; the exact FHA network with both branches, used for Figure 2, differs from Eq. (2.1) by less than 0.1 % around $f_n = 1$ at the values of Q and L_n of this design. At $f_n = 1$ the gain is one at any load: this is the DCX point. Figure 2 shows the gain for three loads with the tank of Section 2.4: between 10 % and 100 % of the load the gain at

$f_n = 0.95$ stays within 0.3 % of one, so the open-loop stage is a stiff DC transformer. The input impedance of the tank is almost resistive at full load (0.2° at 100 %, 7° at 50 %) and becomes inductive at light load (39° at 10 %) because of the magnetizing branch: the bridge current never leads the voltage, which is the condition for ZVS.

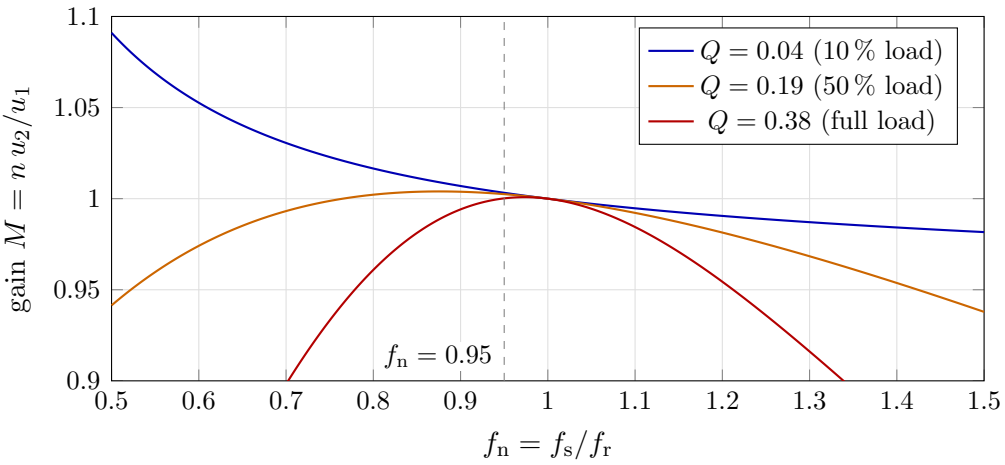


Figure 2: First-harmonic gain of the tank of Section 2.4 ($L_n = 33$) for three loads, exact FHA network with both resonant branches (`figures/fha_gain_50kHz.dat`). The working point $f_n = 0.95$ is marked.

Why f_r a little above f_s. With $f_s < f_r$ the resonant half sine of the current ends before the end of the half period: the rectifying bridge turns off at zero current, and in the remaining interval the driving bridge carries the magnetizing current only, which is what commutates it with ZVS. With $f_r = 52.6$ kHz the half sine lasts 9.5 µs of the 10 µs half period and still ends before the commutation with the tolerances of L_r and C_r ($f_r \pm 7$ % for $C_r \pm 5$ % and $L_r \pm 10$ %). A larger gap would not be a DCX any more: the current becomes more peaked, the conduction losses and the voltage on C_r grow, and the gain starts to move with the load. Above resonance ($f_n > 1$) the rectifier would be turned off while still carrying current, losing the soft turn-off.

2.3. Why the DCX does not regulate

The phase displacement φ between the two bridges, which is the control variable of the dual active bridge [9, 10], is available in the modulator of the model, and the question whether to close a loop on it comes up naturally. With the fundamentals, the power from bridge 1 to bridge 2 is

$$P(\varphi) \simeq \frac{U_1\, n U_2}{2\,|X(f_s)|}\sin\varphi \qquad \text{with} \qquad U_1 = \frac{4}{\pi}u_1, \quad U_2 = \frac{4}{\pi}u_2, \tag{2.2}$$

where $X(f_s) = 2\,(\omega_s L_{r1} - 1/(\omega_s C_{r1}))$ is the series reactance of the whole tank referred to side 1, with L_m neglected. At $f_n = 0.95$ the tank of Section 2.4 gives $X(f_s) = -0.81\,\Omega$, small and capacitive, so that $P \simeq 640\,\text{kW}\,\sin\varphi$: the nominal 50 kW corresponds to $\varphi = 4.5°$ (Figure 3), and the dead time alone is 5.4°. Three facts follow.

1. The stage is extremely stiff: a few degrees of phase move tens of kilowatts, and the gain of a phase-shift loop would be about 13 pu/rad of DC current. Any control on φ needs small gains and the current is then set by the bridge, not by the loop (Section 5.3).
2. Since $X(f_s) < 0$, a positive φ (bridge 2 lagging) would carry power from side 2 to side 1, opposite to the inductive case of the DAB; the sign of the reactance has to enter the control law.
3. When bridge 2 is driven by a phase-shifted PWM instead of following its own current, the rectifier switches with current at the dead time and the tank is forced into capacitive operation: the natural operation of the CLLC near resonance is incompatible with a regulated phase.

The DCX is therefore left in open loop (mode 0 of the control, Section 5.3): bridge 1 at 50 % duty, bridge 2 as a synchronous rectifier driven by the sign of i_{r2}. The power and the direction of the flow are set by the stages around it, as in Figure 1; the voltage of side 2 follows u_1/n minus the drop on an equivalent resistance of about 43 mΩ (Section 2.8). The current and voltage loops of the library block (modes 1 and 2) are kept as an option for other uses of the same stage, not for the SST.

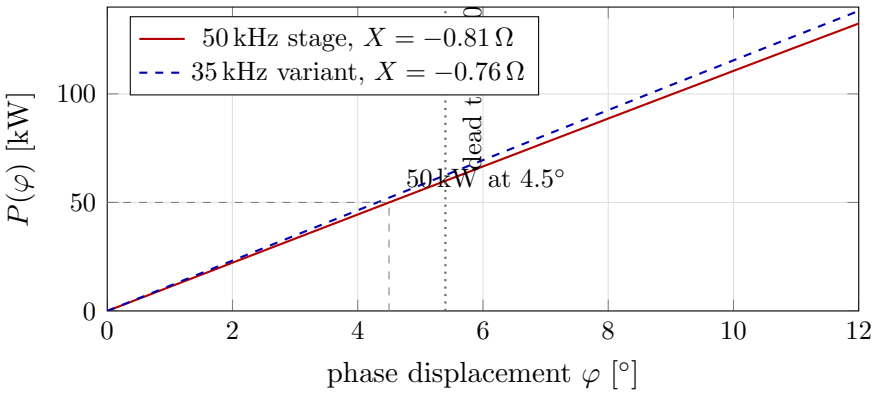
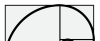


Figure 3: Power against phase displacement of the two bridges with the fundamentals, Eq. (2.2): the nominal power is reached at 4.5°, less than the 5.4° of the dead time at 50 kHz. The stage cannot be regulated through φ with the bridges in PWM.



2.4. Resonant tank and magnetizing inductance

Currents. With $f_s < f_r$ each half period carries the charge $i_{dc1}/(2f_s)$ in a half sine of period $1/f_r$:

$$I_{1pk} = \frac{\pi}{2}\frac{i_{dc1}}{f_n} = 103\,\text{A}, \qquad I_{1rms} = I_{1pk}\sqrt{\frac{f_n}{2}} = 71.2\,\text{A}, \tag{2.3}$$

71.5 A rms and 113 A peak with the magnetizing triangle added. Side 2 carries the same current ($n = 1$).

Magnetizing inductance. The magnetizing current must charge and discharge the output capacitances of one leg of the driving bridge within the dead time. The charge of a leg is $Q_{leg} = 2C_{oss}u_1 = 2 \cdot 0.79\,\text{nF} \cdot 800\,\text{V} = 1.26\,\mu\text{C}$, so the minimum peak is $Q_{leg}/t_d = 4.2\,\text{A}$. With margin, $I_m = 10\,\text{A}$:

$$L_m = \frac{u_1}{4I_m f_s} = \frac{800}{4 \cdot 10 \cdot 50000} = 400\,\mu\text{H}, \qquad t_{ZVS} = \frac{2C_{oss}u_1}{I_m} = 126\,\text{ns} < t_d = 300\,\text{ns}. \tag{2.4}$$

The upper bound for the ZVS is $L_m \le t_d/(8f_sC_{oss}) = 950\,\mu\text{H}$; with $I_m = 10\,\text{A}$ the leg capacitance may grow up to $I_m t_d/u_1 = 3.75\,\text{nF}$ (snubbers, module paralleling) before the ZVS is lost. I_m is 14 % of the resonant peak and its rms value (5.8 A) adds 0.3 A to the winding current; the turn-off of the driving bridge happens at I_m only.

Resonant inductance and capacitance. $L_{r1} = 12\,\mu\text{H}$ on side 1: about 4 µH of transformer leakage and the rest a discrete inductor (Section 2.5). At $f_r = 52.6\,\text{kHz}$,

$$C_{r1} = \frac{1}{(2\pi f_r)^2 L_{r1}} = 0.762\,\mu\text{F}, \qquad Z_0 = \sqrt{\frac{L_{r1}}{C_{r1}}} = 3.97\,\Omega, \tag{2.5}$$

and with $R_{ac} = 10.4\,\Omega$ the tank has $Q = 0.38$ and $L_n = 33$. C_{r1} is a film capacitor for 71 A rms at 50 kHz and 410 V peak ($I_{1pk}/(\omega_r C_{r1})$); the full-bridge drive keeps the DC component off it. The side 2 tank repeats side 1, $L_{r2} = L_{r1}/n^2 = 12\,\mu\text{H}$ and $C_{r2} = n^2 C_{r1} = 0.762\,\mu\text{F}$, so that the resonance is the same in both power directions and both sides see the same currents and voltages. The small Q and the large L_n are typical of a DCX [5]: the tank is not asked to provide gain, only to shape the current into a half sine and to make the magnetizing current available for the ZVS.

Figure 4 shows the idealized waveforms of one side of the tank built from these numbers: the square wave of the bridge, the resonant half sines, the magnetizing triangle and the voltage of the resonant capacitor, which stays constant in the interval between the end of the half sine and the commutation.

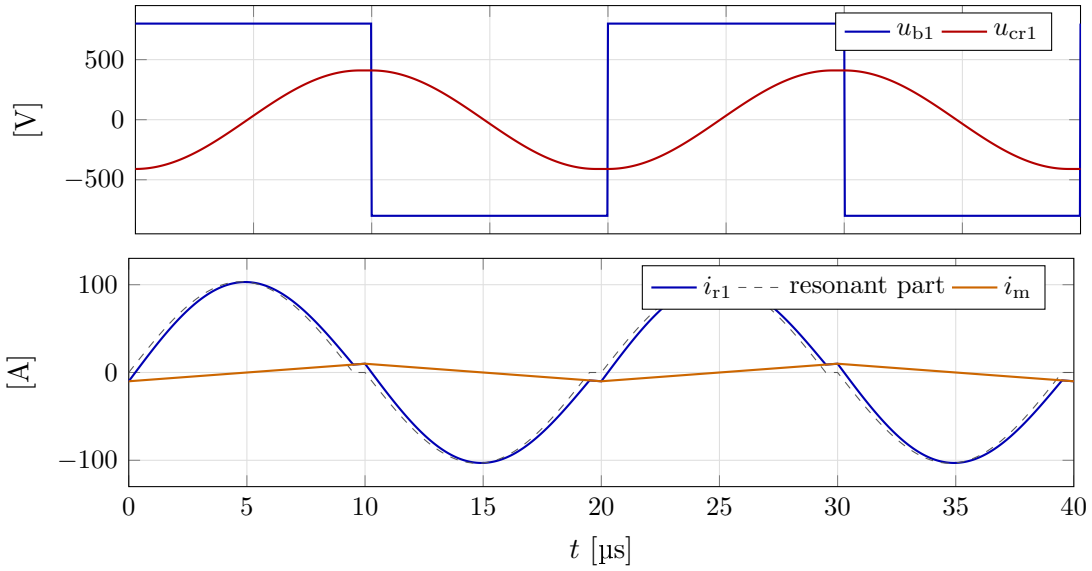


Figure 4: Idealized waveforms of side 1 at full load, two switching periods (`figures/tank_waveforms_50kHz.dat`): bridge voltage and resonant capacitor voltage (top), tank current with its resonant and magnetizing parts (bottom). The half sine ends 0.5 µs before the commutation, where only $I_m = 10$ A is switched.

2.5. Transformer

The transformer is sized by the script `transformer_sizing.m` of the bench folder; Table 2 summarizes the result. Three stacked UU93/152/30 sets of N87-class ferrite give 25.2 cm^2 of core area; eleven turns per winding in one layer of litz wire keep the peak flux density at 0.144 T, set by the core loss and not by the saturation (the first pulse from zero flux reaches 0.29 T, still below saturation, which is why the model can start without a soft start). The gap of about 0.8 mm sets $L_m = 400$ µH; the leakage of about 4 µH (windings on separate layers, no interleaving, 3 mm barrier) is part of L_{r1}, the remaining 10 µH per side is a discrete inductor. Two stacked sets (16 turns) would need two layers per winding and do not fit the 37 mm window; four sets (8 turns) have the same copper and 30 % more core loss.



Table 2: Transformer of the 50 kHz stage (`transformer_sizing.m`).

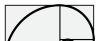
Quantity	**Value**	**Note**
Turns $n_1 : n_2$	11 : 11	one layer of litz per winding
Voltage, current	800 V square wave, 71.5 A rms per winding	113 A peak with I_m
Core	3 stacked UU93/152/30 sets, ferrite N87 class	$A_\mathrm{e} = 25.2\,\mathrm{cm}^2$, $l_\mathrm{e} = 354\,\mathrm{mm}$, 4.5 kg
Peak flux density	0.144 T	set by the core loss; 0.29 T on the first pulse from zero flux
Air gap	about 0.8 mm in total	for $L_\mathrm{m} = 400\,\mu\mathrm{H}$ (2.4 mH without gap)
Windings	litz 3000×0.1 mm, $24\,\mathrm{mm}^2$ ($J = 3\,\mathrm{A\,mm^{-2}}$), 8.2 mm bundle	11 turns in one layer of 90 mm, radial build 21 mm of the 37 mm window
Insulation	3 mm barrier and shield between the windings	800 V DC isolation, bobbin 2 mm
Resistance	3.2 mΩ DC, about 4.7 mΩ AC per winding ($F_\mathrm{r} = 1.5$)	model $R_\mathrm{r1} = R_\mathrm{r2} = 6\,\mathrm{m\Omega}$ with terminations and the discrete inductor
Leakage	about 4 µH in total	the rest of $L_\mathrm{r1} + L_\mathrm{r2} = 24\,\mu\mathrm{H}$ is discrete, about 10 µH per side
Losses	about 62 W core and 48 W copper	0.22 % of 50 kW; model $R_\mathrm{fe} = u_1^2/60\,\mathrm{W} = 10.7\,\mathrm{k\Omega}$

The Steinmetz law of the script is anchored to one datasheet point of the material ($49.5\,\mathrm{kW\,m^{-3}}$ at 35 kHz and 0.156 T, 100 °C) with exponents 1.5 and 2.5, and the AC resistance factor of the litz winding is taken as $F_\mathrm{r} = 1.5$: both are assumptions to replace with the datasheet of the chosen material and wire.

2.6. Semiconductors and synchronous rectifier drive

Both bridges are full bridges of 1200 V SiC MOSFET half-bridge modules, the same device on both sides because both DC links are at 800 V. The library dataset `wolfspeed_SiC_MOSFET_CAB006M12GM3` is used: 6.6 mΩ at 100 °C, $C_\mathrm{oss} = 0.79\,\mathrm{nF}$ (the value of the ZVS sizing above). The module has margin at 71.5 A rms and 113 A peak; a module of 13 mΩ to 16 mΩ is the cost option, with 70 W to 100 W more per bridge.

- Conduction: $2R_\mathrm{ds,on}I_\mathrm{1rms}^2 = 68\,\mathrm{W}$ per bridge (two devices in series in the current path).



- Switching of the driving bridge: ZVS turn-on at any load, turn-off at $I_{\mathrm{m}} = 10\,\mathrm{A}$ only, about 2 W per switch with the datasheet E_{off} scaled to 10 A and 800 V.
- The rectifying bridge switches at almost zero current: gate and body-diode losses only.

Synchronous rectifier drive. The rectifying bridge must follow the sign of its own current and not the PWM of the other bridge: the dead time is already 5.4° at 50 kHz, and Section 2.3 showed that a few degrees move tens of kilowatts. The SR logic of the bench (blocks `sr_gate_logic` for bridge 2 and `sr1_gate_logic` for bridge 1) and of the library block (Section 5.3) turns a diagonal on when $|i| > i_{\mathrm{on}} = 5\,\mathrm{A}$ (0.08 pu) and off when $|i| < i_{\mathrm{off}} = 1.9\,\mathrm{A}$ (0.03 pu), with the state held in between. At the zero crossing the current slope is $I_{1\mathrm{pk}}\,\omega_{\mathrm{r}} = 34\,\mathrm{A\,\mu s^{-1}}$, so the switch turns on 146 ns after the zero and off 55 ns before it, with the body diodes conducting in between. When the power flows from side 2 to side 1 the roles of the two bridges swap, in the bench with `param.sim.sink` and in the library block with `p.ctrl.sr_mode_1` and `p.ctrl.sr_mode_2`.

2.7. DC links and chokes

The rectified current is a train of half sines. Its AC part is $\sqrt{I_{1\mathrm{rms}}^2 - i_{\mathrm{dc}}^2} = 34\,\mathrm{A}$ rms at $2f_{\mathrm{s}} = 100\,\mathrm{kHz}$, and the charge of each half period above the mean is $0.105\,i_{\mathrm{dc}}/f_{\mathrm{s}} = 132\,\mu\mathrm{C}$. With $C_{\mathrm{dc}} = 500\,\mu\mathrm{F}$ per DC link the ripple is 0.26 V peak to peak and the loss on $1\,\mathrm{m\Omega}$ of ESR is about 1 W per link: the capacitors are set by their ripple current rating, not by the ripple voltage, which calls for a film bank. In the SST module the DC link of side 1 is shared with the single-phase inverter and is sized on the 100 Hz ripple of the inverter (3 mF in the bench of [1]); 500 µF is the value of the stand-alone bench. The DC chokes ($L_{\mathrm{dc}} = 250\,\mu\mathrm{H}$, $2\,\mathrm{m\Omega}$, 8 W each at 62.5 A) resonate with 500 µF at 450 Hz; the current-source load of the bench ramps in 20 ms, well above that period.

2.8. Loss budget

Table 3 is the sizing estimate of the losses at 50 kW. Seen from side 2, the DCX is a voltage source u_1/n with the series resistance $R_{\mathrm{eq}} \simeq 4R_{\mathrm{ds,on}} + R_{\mathrm{r1}} + R_{\mathrm{r2}} + 2R_{\mathrm{Ldc}} \simeq 43\,\mathrm{m\Omega}$, that is 2.7 V at 62.5 A; the model adds the numerical resistance of Section 4.4 ($125\,\mathrm{m\Omega}$ at $t_{\mathrm{c}} = 50\,\mathrm{ns}$), so that the efficiency between the DC links of a run is expected around 98 % and the script `power_loss_calculus.m` separates the two.



Table 3: Loss budget of the 50 kHz stage at 50 kW (sizing estimate).

Item	**Loss**	**Basis**
Bridge 1, conduction	68 W	$2 \times 6.6\,\mathrm{m\Omega} \times (71.5\,\mathrm{A})^2$
Bridge 1, switching and gate	15 W	ZVS turn-on, turn-off at I_m
Resonant inductors and $C_{\mathrm{r}1}$, $C_{\mathrm{r}2}$	40 W	about 20 W per side
Transformer	110 W	Table 2
Bridge 2, conduction	68 W	as bridge 1 (SR in load, driving bridge in sink)
Bridge 2, gate and body diodes	15 W	
DC chokes and DC-link ESR	18 W	$2 \times 8\,\mathrm{W} + 2 \times 1\,\mathrm{W}$
Auxiliaries	30 W	gate drivers, control, fans
Total	**365** W	$\eta \simeq 99.3\,\%$

3. The 35 kHz variant, 790 V to 700 V

3.1. Origin and purpose

The repository keeps a second bench folder, `sst_cllc_dcx_790V_700V_50kW`, and the library a second parameter set, `cllc_dcx_param_example_35kHz.m`. Its numbers are not a second requirement of the SST: 790 V on side 1 and 700 V on side 2 are what was read on the first runs of the 800 V / 800 V model with the local solver step at $t_\mathrm{c} = 500\,\mathrm{ns}$, where the numerical damping of the Backward Euler method (Section 4.4) adds about $1.25\,\Omega$ in series with the tank and drops some 80 V at 62.5 A. The reading was kept as a second design case, with the switching frequency lowered to 35 kHz where the same step costs less damping, and both benches are maintained so that the two frequencies can be compared on the same semiconductors and the same core family. The specification of the module remains the 50 kHz stage of Section 2.

3.2. Sizing

The design follows the same steps with the same rules: $f_\mathrm{n} = 0.95$, $I_\mathrm{m} = 10\,\mathrm{A}$ for the ZVS of the driving bridge within 300 ns, symmetrical tank, one layer of litz per winding. Table 4 puts the two parameter sets side by side. With 9 : 8 turns the no-load voltage of side 2 is 702 V; the ratio 17 : 15 would give 697 V but needs two layers. At 35 kHz the volt-seconds per half period are 43 % more than at 50 kHz, so $n_1 A_\mathrm{e}$ grows by the same factor: nine turns of 8.3 mm litz are the most that fit in one layer and five stacked UU93/152/30 sets are needed for 0.15 T (7.5 kg against 4.5 kg). Four sets (0.187 T) save 1.5 kg with



92 W of core loss instead of 66 W and a first pulse at 0.37 T, close to saturation without a soft start; three sets need 18 : 16 in two layers per winding, which fit the window only at 4 A mm^{-2}.

The loss budget of the variant, on the same basis as Table 3, is about 400 W ($\eta \simeq 99.2\,\%$): the transformer takes about 122 W instead of 110 W and bridge 2 conducts 81 A rms instead of 71.5 A (87 W instead of 68 W), while the lower frequency saves little, because the driving bridge already switches with ZVS and the rectifier at zero current. The tank reactance at the switching frequency is $X(f_\mathrm{s}) = -0.76\,\Omega$ and $P \simeq 665\,\mathrm{kW}\,\sin\varphi$, 50 kW at 4.3° (Figure 3): the same conclusions of Section 2.3 apply.



Table 4: The two parameter sets side by side (library files `cllc_dcx_param_example.m` and `cllc_dcx_param_example_35kHz.m`).

Quantity	**50** kHz **stage (specification)**	**35** kHz **variant**
DC voltages u_1 / u_2	800 V / 800 V	790 V / 700 V (702 V at no load)
Turns $n_1 : n_2$	11 : 11	9 : 8
DC currents i_{dc1} / i_{dc2}	62.5 A / 62.5 A	63.3 A / 71.4 A
f_s / f_r	50 kHz / 52.6 kHz	35 kHz / 36.8 kHz
L_{r1} / C_{r1}	12 µH / 0.762 µF	16 µH / 1.17 µF
L_{r2} / C_{r2}	12 µH / 0.762 µF	12.6 µH / 1.48 µF
L_m (side 1), L_n	400 µH, 33	560 µH, 35
Z_0, R_{ac}, Q	3.97 Ω, 10.4 Ω, 0.38	3.70 Ω, 10.1 Ω, 0.37
Series resistances R_{r1} / R_{r2}	6 mΩ / 6 mΩ	7 mΩ / 5.5 mΩ
Core-loss resistance R_{fe}	10.7 kΩ (60 W)	9.5 kΩ (66 W)
Tank current side 1, rms / peak	71.2 A / 103 A (113 A with I_m)	72.4 A / 105 A (115 A with I_m)
Tank current side 2, rms / peak	as side 1	81.1 A / 118 A
C_{r1} / C_{r2} peak voltage	410 V / 410 V	390 V / 345 V
Magnetizing current peak, ZVS time	10 A, 126 ns	10 A, 124 ns
Transformer core	3 stacked UU93/152/30, 0.144 T, 4.5 kg	5 stacked UU93/152/30, 0.149 T, 7.5 kg
Transformer losses	about 110 W	about 122 W
Loss budget, efficiency	365 W, 99.3 %	about 400 W, 99.2 %
$X(f_s)$, $P(\varphi)$	$-0.81\,\Omega$, $637\,\text{kW}\sin\varphi$ (4.5°)	$-0.76\,\Omega$, $665\,\text{kW}\sin\varphi$ (4.3°)
DC links, chokes, devices	500 µF / 1 mΩ, 250 µH / 2 mΩ, CAB006M12GM3	the same
SR thresholds on / off	0.08 pu / 0.03 pu (5 A / 1.9 A)	0.08 pu / 0.03 pu (5.1 A / 1.9 A)
R_{num} per unit of step	$2.50\,\Omega\,\mu\text{s}^{-1}$	$1.63\,\Omega\,\mu\text{s}^{-1}$

A bench of the specification at 35 kHz, 800 V to 800 V on the same tank, is obtained from the variant with $n_2 = 9$ and $u_2 = 800$ V in its parameter file: the tank of the variant does not depend on the ratio, only the side 2 components do.

4. Simulation model in solid-state-transformers

4.1. The bench

Each bench folder (`sst_cllc_dcx_800V_800V_50kW` with `sst_cllc_dcx.slx`, `sst_cllc_dcx_790V_700V_50kW` with `sst_cllc_dcx_35kHz.slx`) holds one Simscape model built with the discrete blocks of the library: full bridges with the MOSFET model `mos_simple_2th`, the CLLC transformer object of `hw_cllc_single_phase_setup`, the modulator `dab_single_phase_pwm_modulator` and the battery `lithium_ion_battery`. Figure 5 is the bench in load mode. The root of the model has `battery_1` (the 800 V source of side 1, with the capacitor at its terminals), the DCX `dcdc_cllc_modA`, the `global timing` block (local clock `time_cllc_module_1` and `time_master`) and the `load`, a controlled current source on side 2 that ramps to 62.5 A in 20 ms. Inside the DCX the two DC chokes `LFi_dc1/2`, the DC-link capacitors `CFi_dc1/2`, the two bridges and the tank; the blocks `sr1_gate_logic` and `sr_gate_logic` in front of the two bridges select, with `param.cllc.sr1.mode` and `param.cllc.sr.mode`, between the modulator (mode 0), the synchronous rectifier on the sign of the bridge current (mode 1) and the diode rectifier with all gates off (mode 2, reference case). The subsystem `input_current_control` holds the modulator with the phase displacement, which in the DCX use stays at zero.

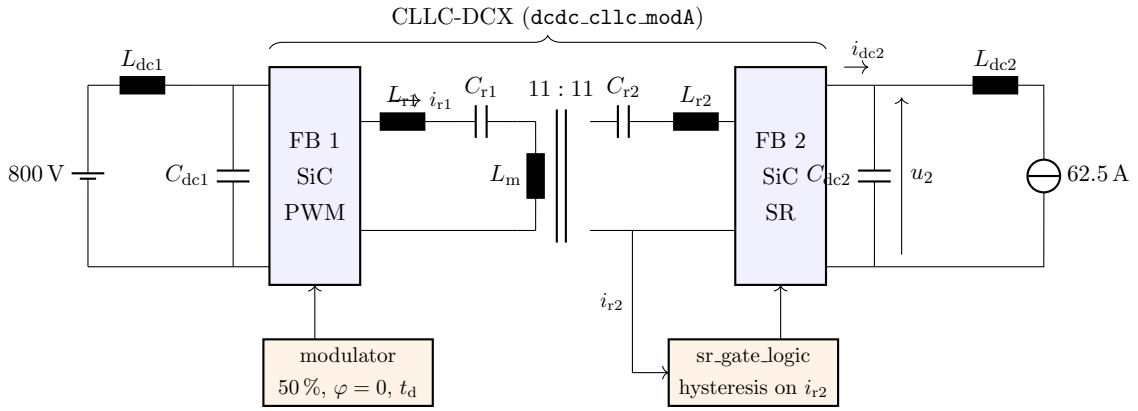


Figure 5: Simulation bench of the DCX in load mode: battery on side 1 behind the DC choke and the DC-link capacitor, CLLC with the two-sided tank, bridge 2 driven as synchronous rectifier by the sign of i_{r2}, DC link 2 and current source behind L_{dc2} (the load).

4.2. Power flow and bench modes

The direction of the power is chosen with `param.sim.sink` in `init_model.m` (Table 5). In load mode the battery feeds side 1 and the current source draws 62.5 A from side 2; in the two sink modes the power flows from side 2 to side 1, either through the same current source pushing −62.5 A into side 2 (the downstream stage working in sink, the



battery being charged) or on a mirror bench where a stiff voltage source U_2 replaces the current source and a resistor R_1 replaces the battery (the blocks are swapped by hand in the model). In both sink modes `init_model.m` swaps the two SR modes and sets the initial voltages, so that the modulator drives bridge 2 at 50 % duty and bridge 1 rectifies. The expected result of the sink runs is the same current and the same R_{eq} as in load: this is the symmetry of the CLLC that the sizing relies on.

Table 5: Bench modes of the DCX (`param.sim.sink`).

Mode	Bench	Bridge 1	Bridge 2
0, load	battery on side 1, current source +62.5 A on side 2 (ramp in 20 ms)	modulator (`sr1.mode` 0)	SR on i_{r2} (`sr.mode` 1)
1, sink	battery on side 1 (charged), current source −62.5 A on side 2	SR on i_{r1} (`sr1.mode` 1)	modulator (`sr.mode` 0)
2, mirror	resistor R_1 on side 1, voltage source U_2 on side 2, set so that $u_1 = 800\,\mathrm{V}$ with R_{eq}	SR on i_{r1}	modulator

4.3. Switch model

The switches of the bench are the library model `mos_simple_2th`: a MOSFET with two threshold voltages, the conduction resistance of the dataset and the antiparallel body diode, without switching dynamics. The output capacitance C_{oss} is optional: with `param.sim.use_coss = 1` the capacitance of the dataset is placed across each switch, with a series resistance of 10 ns time constant for the numerical damping, so that the real dead-time transitions (the 126 ns of Eq. (2.4)) and the ringing of the rectifier nodes in the zero-current interval can be seen. Without it the commutations are ideal and the run is faster; the losses of the switches are then the conduction losses only, plus the gate and switching terms added by the post-processing scripts from the dataset. The library block of Section 5 has the same two options (`p.hw.coss_on`, `p.hw.Coss`, `p.hw.tau_coss`).

4.4. Numerical settings and the damping of the fixed-step solver

The Simscape network runs on a local solver with the Backward Euler method and the fixed step t_{c}, which is also the sample time of the control. Backward Euler is dissipative: at the angular frequency ω each inductor gets about $\omega^2 L t_{\mathrm{c}}/2$ of extra series resistance and each capacitor about $t_{\mathrm{c}}/(2C)$. For the two sides of the tank, referred to side 1,

$$R_{\mathrm{num}} \simeq 2\left(\frac{\omega_{\mathrm{s}}^2 L_{\mathrm{r1}}\, t_{\mathrm{c}}}{2} + \frac{t_{\mathrm{c}}}{2C_{\mathrm{r1}}}\right) = t_{\mathrm{c}}\left(\omega_{\mathrm{s}}^2 L_{\mathrm{r1}} + \frac{1}{C_{\mathrm{r1}}}\right), \tag{4.1}$$



that is 2.50 Ω per microsecond of step for the 50 kHz stage and 1.63 Ω for the 35 kHz variant. This resistance is in series with the 43 mΩ of the physical R_{eq} (Section 2.8) and dissipates $R_{num} I_{1rms}^2$ in the run (Table 6 and Figure 6). The relative weight of R_{num} grows with $\omega_s Q t_c$, so the 50 kHz stage needs a shorter step than the variant for the same accuracy. The two other choices are worse for this circuit: the trapezoidal method has no damping but rings on the switching edges, where the ZVS is read; the Partitioning solver chatters on the SR bridge at every zero of the current. The step must also resolve the dead time (300 ns) and the 126 ns ZVS transition when C_{oss} is on: 50 ns is the value of the bench for the voltages, the losses and the ZVS; 500 ns is good for structural checks only. The scripts of the bench print R_{num} and the power it dissipates for the step of the run; halving t_c halves it and doubles the run time.

Table 6: Numerical damping of the Backward Euler local solver, Eq. (4.1), at the full-load tank current (71.2 A rms at 50 kHz, 72.1 A rms at 35 kHz).

	50 kHz **stage**		**35** kHz **variant**		
Step t_c	R_{num}	loss	R_{num}	loss	**Use**
25 ns	0.062 Ω	0.32 kW	0.041 Ω	0.21 kW	accuracy runs
50 ns	0.125 Ω	0.63 kW	0.082 Ω	0.42 kW	voltages, losses, ZVS (bench default)
100 ns	0.25 Ω	1.3 kW	0.16 Ω	0.85 kW	SST bench on `dev`
500 ns	1.25 Ω	6.3 kW	0.82 Ω	4.2 kW	structural checks; origin of the 790 V to 700 V reading
1 µs	2.5 Ω	12.7 kW	1.63 Ω	8.5 kW	quick runs of the SST bench, not for the DCX figures

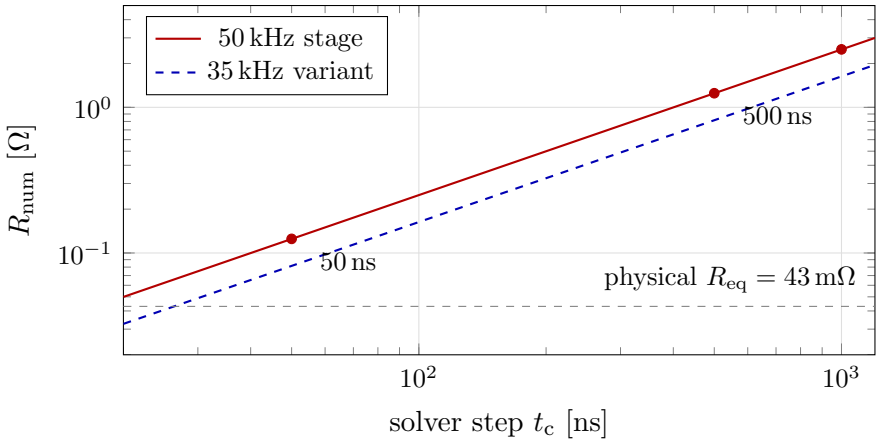


Figure 6: Numerical series resistance of the tank against the solver step, Eq. (4.1), compared with the physical equivalent resistance of the stage. At 500 ns the numerical resistance is thirty times the physical one; at 50 ns it is still three times.



State of the bench files. At the time of writing the parameter file of the 800 V / 800 V bench (`private/cllc_dcx_param.m`) carries the switching frequency as a pair of lines, 50 kHz commented out and 35 kHz active, with the resonant capacitor derived from $f_r = f_s/0.95$: in that state the 800 V bench runs at 35 kHz with $L_{r1} = 12\,\mu H$ and $C_{r1} = 1.56\,\mu F$, which is a third tank, not the one of Table 4. The sizing of this note refers to the 50 kHz line; the 35 kHz runs belong to the variant folder.

4.5. Checks to do with the model

No run of the DCX is saved in the repository yet. Table 7 lists the first runs and what each has to be compared with; the scripts named in the last column are those of the bench folder (`power_loss_calculus.m` checks a run against the sizing, `plotting_results.m` draws the figures, `cllc_analysis.m` computes the FHA gain, the input impedance and $P(\varphi)$).

Table 7: Checks to do with the model against the sizing.

Item	Sizing	How
Tank currents i_{r1}, i_{r2} rms / peak	71.5 A / 113 A	power_loss_calculus.m
Magnetizing current peak	10 A	figure 4 of plotting_results.m
Gain and phase at full load	$u_2 = u_1/n$, i_{r1} in phase with u_{b1}	power_loss_calculus.m
ZVS of the driving bridge	126 ns of 300 ns	figure 7, zvs_check, with use_coss = 1
Bridge losses	68 W + 15 W per bridge	power_loss_calculus.m
C_{r1} peak voltage	410 V	from the peak current
DC link 2 ripple and capacitor current	0.26 V p-p, 34 A rms	figures 4 and 6
R_{eq} seen from side 2	43 mΩ + 125 mΩ numerical at 50 ns	power_loss_calculus.m
Efficiency between the DC links	99.3 % physical, about 98 % with R_{num}	power_loss_calculus.m
Sink mode, both benches	same currents and R_{eq} as load	param.sim.sink = 1, 2
Light load, 10 % and 50 %	gain within 0.3 %, SR thresholds, ZVS with the same I_m	param.cllc.sink.I2
Start-up	peak of i_{r1} and of u_2 without soft start	figure 3
Tank tolerances	$f_r \pm 7$ % ($C_r \pm 5$ %, $L_r \pm 10$ %)	two runs at the corners

5. The library block cllc_dcx

5.1. Motivation and structure

The SST bench of [1] puts two DCX and two inverters in one model. Built with the discrete blocks of Figure 5, each DCX costs several hundred nonvirtual blocks, and the MATLAB Home licence used for the project limits a model to 1000 nonvirtual blocks, referenced models and library links included. The block cllc_dcx/cllc_dcx_mod1 of pwr-control/library (file masked_models/cllc_dcx/cllc_dcx.slx) packs one complete stage into two nonvirtual blocks (Figure 7):

- the power stage is one Simscape language component, cllc_ssc.cllc_dcx_single_phase (source foundation/pwr_electronics/+cllc_ssc/cllc_dcx_single

_phase.ssc, compiled into cllc_ssc_lib.slx with ssc_build cllc_ssc): eight numerical switches with antiparallel diodes and optional C_{oss}, the two-sided tank, the transformer with L_m and R_{fe}, and optional DC-link capacitors and DC chokes on both sides;

- the modulator of both bridges, the synchronous-rectifier logic of both bridges and the control are one MATLAB Function, cllc_dcx_ctrl_pwm (source masked_models/cllc_dcx/private/cllc_dcx_ctrl_pwm.m), discrete at the sample time p.pwm.tc, which is the step of the Simscape local solver.

Around them, the block has only the converters between the two domains: gates (Simulink-PS, the eight gate voltages), meas (PS-Simulink, the fifteen measurements in pu) and meas_delay, a Unit Delay of one t_c that breaks the algebraic loop between measurements and gates. The script update_cllc_dcx.m builds the block from the two sources and refreshes it after an edit; the mask of the block carries the single parameter structure p.

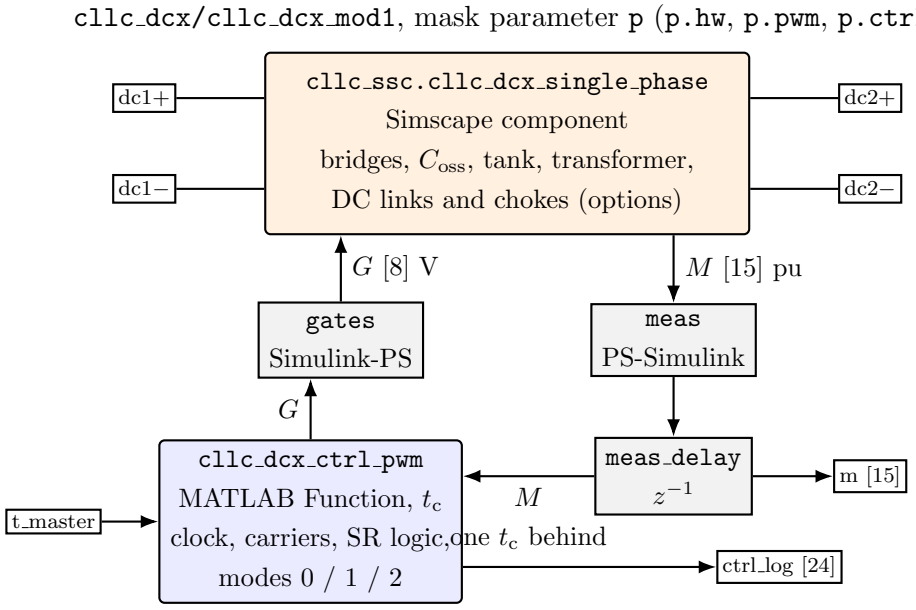


Figure 7: Structure of the library block cllc_dcx_mod1: one Simscape component for the power stage, one MATLAB Function for modulator, synchronous rectifiers and control, and the converters between the two domains with the one-step delay of the measurements.

5.2. Ports, measurements and gates

The electrical ports are dc1+, dc1− on the left (side 1) and dc2+, dc2− on the right (side 2); the two sides are galvanically separate inside the block, so the model needs one Electrical Reference per side. The Simulink input is t_master, the master time of the



model (the Clock, or the master time of the models with a global timing block); the outputs are `m`, the measurement vector M in pu, and `ctrl_log`, the control log. Table 8 lists the fifteen measurements with their sign conventions and Table 9 the order of the eight gate commands; the bases of the pu system are `p.hw.udc_nom` and `p.hw.idc_nom` (800 V and 62.5 A for the 50 kHz stage).

Table 8: Measurement vector M (output `m`, 15 pu values, one t_c behind at the input of the control).

Index	Signal	Meaning and sign
1	u_{dc1}	terminal voltage of side 1
2	i_{dc1}	terminal current of side 1, positive into `dc1+`
3	u_{dc2}	terminal voltage of side 2
4	i_{dc2}	terminal current of side 2, positive out of `dc2+`
5	i_{r1}	tank current of side 1, out of midpoint 1a into the tank
6	i_{r2}	tank current of side 2, out of the secondary into midpoint 2a
7, 8	u_{b1}, u_{b2}	output voltages of the two bridges
9, 10	u_{cr1}, u_{cr2}	voltages of the two resonant capacitors
11, 12	u_{bus1}, u_{bus2}	DC bus voltages of the two bridges (behind the chokes)
13	i_m	magnetizing current, referred to side 1
14, 15	i_{c1}, i_{c2}	DC-link capacitor currents, positive from (+) to (−)

Table 9: Gate vector G (eight voltages, `p.pwm.gate_on` / `p.pwm.gate_off` = 18 V / −3 V against V_{th} = 2.5 V).

Index	Switch	Position
1, 2	1a+, 1a−	bridge 1, leg a, upper and lower switch (midpoint 1a feeds the tank)
3, 4	1b+, 1b−	bridge 1, leg b
5, 6	2a+, 2a−	bridge 2, leg a (midpoint 2a receives the secondary)
7, 8	2b+, 2b−	bridge 2, leg b

The bridge states of the modulator are +1 when the diagonal (a+, b−) is on, −1 when the diagonal (a−, b+) is on and 0 during the dead time or when both are off; the diagonal (a+, b−) applies $+u_{dc}$ to the tank.

5.3. Control chain

The MATLAB Function reproduces, in one block, the chain of the bench model (names of the bench blocks in brackets). Every step t_c:

- **local clock** [`global timing`]: $t_{loc} = t_{master}\, k_{time}\, (1 + \mathrm{corr})$, with the optional alignment PI of the local clock on $t_{master} - t_{loc}$ (gains zero in the DCX: free-running clock equal to the master time);
- **carrier 1** [`modulator`]: triangle 0..1 at f_{pwm} from the local clock, with the optional soft start that sweeps the frequency from `p.pwm.f_start` down to f_{pwm} in `p.pwm.t_soft`; the control trigger TRGO is raised at the carrier bottom, and also at the top with `double_update`;
- **carrier 2**: carrier 1 delayed by φ (bridge 2 lags bridge 1 for $\varphi > 0$); φ changes at the trigger, where carrier 2 is far from the switching level for $|\varphi| < \pi/2$;
- **acquisitions** [`acquisitions`]: average of u_{dc1}, i_{dc1}, u_{dc2}, i_{dc2}, i_{c1}, i_{c2} over the control period, quantized with `adc_q` at the trigger (the $2f_{pwm}$ ripple of the DC quantities averages out, as with an anti-aliasing filter or ADC oversampling);
- **gates**: a bridge in PWM (`sr_mode` 0) has the diagonal (a+, b−) on while its carrier is above $1/2 + t_d f$ and the diagonal (a−, b+) on while it is below $1/2 - t_d f$ (50 % duty, dead time around both edges); a bridge as synchronous rectifier (`sr_mode` 1) [`sr_gate_logic`, `sr1_gate_logic`] has the diagonal (a+, b−) on while the current entering its midpoint a is above `sr_i_on`, the diagonal (a−, b+) while it is below −`sr_i_on`, both off once $|i| <$ `sr_i_off`, held in between (hysteresis); `sr_mode` 2 keeps all gates off (diode rectifier). The current entering midpoint 2a is $i_{r2} = M(6)$, the one entering midpoint 1a is $-i_{r1} = -M(5)$.

On the rising edge of TRGO (no trigger on the first evaluation, as the Trigger block of the bench):

- **mode 0**, the design use of the stage: $\varphi =$ `p.ctrl.phi0`, open-loop DC transformer with the rectifying bridge as synchronous rectifier; `phi0` is zero also when both bridges are in PWM;
- **mode 1**, current loop: PI of $i_{ref} - i_{meas}$ to φ, limited to ±`phi_max`, clip flag and integrator release while clipped (as the dq vector PI of the inverter blocks);
- **mode 2**, voltage loop: PI of $u_{ref} - u_{meas}$ to i_{ref}, limited to ±`i_lim`, then the current loop of mode 1 (cascade).

The regulated quantities are those of side 2 by default (`p.ctrl.side` = 2): $u_{meas} = u_{dc2}$ and $i_{meas} = i_{dc2} + i_{c2}$, the current that bridge 2 rectifies into DC link 2 (terminal current plus capacitor current, because the terminal current alone is set by the load and not by the converter); with `side` = 1, $u_{meas} = u_{dc1}$ and $i_{meas} = i_{c1} - i_{dc1}$. The loops command $\varphi = \varphi_{sign}\, p_{dir}\, \varphi_{PI}$, where $p_{dir} = +1$ for side 2 regulated and −1 for side 1, and $\varphi_{sign} = \mathrm{sign}\, X(f_s)$ is computed from `p.hw` (both tanks referred to side 1, L_m neglected) unless



`p.ctrl.phi_sign` is given: this is where point 2 of Section 2.3 enters. The gains of the example files (`kp_i` = 0.02 rad/pu, `ki_i` = 20 rad/(pu s), `kp_u` = 1, `ki_u` = 50 1/s, `i_lim` = 1.2 pu) are conservative starting values: with a plant gain of about 13 pu/rad the current loop has a loop gain of 0.25 and an integral corner at 1000 rad/s, the voltage loop a crossover at about 25 Hz on the 6.4 ms time constant of DC link 2. They are to be tuned on the model if the loops are ever used; mode 0 is the normal use.

Table 10 lists the 24 signals of the control log; the function `cllc_dcx_log_index.m` returns their indices by name, so that a plot script reads `idx.phi` instead of a number.



Table 10: Control log ctrl_log (24 signals, indices from cllc_dcx_log_index.m). The first ten and the last two are updated every step, the others at the control trigger and held in between.

Index	Name	Meaning
1	t_loc	local time of the block [s]
2	ramp1	carrier of bridge 1, triangle 0..1
3	trgo	control trigger flag
4	corr	correction of the local clock (alignment PI)
5	f_c	switching frequency of this step [Hz], sweeps during the soft start
6	phi	phase of bridge 2 after bridge 1 [rad]
7	ramp2	carrier of bridge 2
8	state_1	bridge 1 state: +1 diagonal (1a+, 1b−), −1 diagonal (1a−, 1b+), 0 both off
9	state_2	bridge 2 state, same convention
10	u_dc1	acquired (period average, quantized) terminal voltage of side 1 [pu]
11	i_dc1	acquired terminal current of side 1, positive into dc1+ [pu]
12	u_dc2	acquired terminal voltage of side 2 [pu]
13	i_dc2	acquired terminal current of side 2, positive out of dc2+ [pu]
14	u_meas	regulated voltage (u_{dc2}, or u_{dc1} with side = 1) [pu]
15	i_meas	regulated current, rectified by the bridge into the DC link of the regulated side [pu]
16	u_ref	voltage reference (mode 2, else 0) [pu]
17	i_ref	current reference (mode 1: p.ctrl.i_ref; mode 2: output of the voltage PI) [pu]
18	clip_u	clip flag of the voltage PI
19	phi_pi	output of the current PI, before the sign of the reactance and of the side [rad]
20	clip_i	clip flag of the current PI
21	mode	p.ctrl.mode
22	enable	gates enabled ($t_{master} \geq$ p.pwm.t_enable)
23	i_r1	tank current of side 1 ($M(5)$, one t_c behind) [pu]
24	i_r2	tank current of side 2 ($M(6)$, one t_c behind) [pu]

5.4. Parameter structure

The mask of the block takes one structure p with three fields. p.hw holds the parameters of the Simscape component with the same names and units; the five options (coss_on, cap_dc1, ind_dc1, cap_dc2, ind_dc2) are 0/1 in the structure and the mask turns them into the enumeration cllc_ssc.onoff. p.pwm holds the carriers and the gates, p.ctrl



the control. Table 11 lists the fields with the values of the two example files; the switches of the component are numerical (threshold, on-state resistance, off-state conductance), so R_{on} is the conduction drop of the module and the switching losses are not in the model.

Table 11: Parameter structure `p` of the block, with the values of `cllc_dcx_param_example.m` (50 kHz) and `cllc_dcx_param_example_35kHz.m`.

Field	Meaning	**50** kHz	**35** kHz
`p.hw` – power stage (`cllc_ssc.cllc_dcx_single_phase`)			
`Vth`, `Ron`, `Goff`	switch threshold, on resistance, off conductance	2.5 V, 6 mΩ, $1 \times 10^{-6}\,\Omega^{-1}$	the same
`coss_on`, `Coss`, `tau_coss`	C_{oss} across each switch, value, time constant of its series resistance	1, 0.79 nF, 10 ns	the same
`cap_dc1`, `Cdc1`, `Rdc1`, `Vdc10`	DC-link capacitor 1, capacitance, ESR, initial voltage	1, 500 µF, 1 mΩ, 800 V	1, 500 µF, 1 mΩ, 790 V
`ind_dc1`, `Ldc1`, `RLdc1`	DC choke 1, inductance, resistance	1, 250 µH, 2 mΩ	the same
`Lr1`, `Rr1`, `Cr1`	tank of side 1	12 µH, 6 mΩ, 0.762 µF	16 µH, 7 mΩ, 1.17 µF
`n1`, `n2`, `Lm`, `Rfe`	transformer turns, magnetizing inductance and core-loss resistance (side 1)	11, 11, 400 µH, 10.7 kΩ	9, 8, 560 µH, 9.5 kΩ
`Lr2`, `Rr2`, `Cr2`	tank of side 2 (L_{r1}/n^2, R_{r1}/n^2, $n^2 C_{r1}$)	12 µH, 6 mΩ, 0.762 µF	12.6 µH, 5.5 mΩ, 1.48 µF
`cap_dc2`, `Cdc2`, `Rdc2`, `Vdc20`	DC-link capacitor 2	1, 500 µF, 1 mΩ, 800 V	1, 500 µF, 1 mΩ, 700 V
`ind_dc2`, `Ldc2`, `RLdc2`	DC choke 2	1, 250 µH, 2 mΩ	the same
`udc_nom`, `idc_nom`	bases of the pu system of M	800 V, 62.5 A	790 V, 63.3 A
`p.pwm` – carriers and gates			
`tc`	sample time of the block = local solver step	50 ns (argument)	the same
`fpwm`, `phase`, `double_update`	switching frequency, phase of carrier 1, trigger at both carrier ends	50 kHz, 0, 0	35 kHz, 0, 0
`dead_time`	dead time of the bridges in PWM	300 ns	the same
`gate_on`, `gate_off`	gate levels	18 V, −3 V	the same
`t_enable`	gates enabled from this master time	0	the same
`f_start`, `t_soft`	optional soft start: initial frequency and sweep time (both or none)	not set	not set
`p.ctrl` – control			
`ts`	control period ($1/f_{pwm}$, $1/(2f_{pwm})$ with `double_update`)	20 µs	28.6 µs
`k_time`, `kp_align`, `ki_align`, `lim_up_align`, `lim_down_align`	local clock ratio and alignment PI	1, 0, 0, 0, 0	the same
`adcq`	quantization of the acquisitions	1/4096	the same
`mode`, `phi0`	0 open-loop DCX, 1 current	0, 0	the same

5.5. Build and use

The component is compiled once with `ssc_build cllc_ssc` from `foundation/pwr_electronics`, which produces `cllc_ssc_lib.slx`; `update_cllc_dcx.m` then loads the MATLAB Function source into the block of `cllc_dcx.slx` and is the script to run after any edit of `cllc_dcx_ctrl_pwm.m`. In a model, the block is dropped from the library, the two sides are connected to their Electrical References, `t_master` to the Clock or to the master time of the model and the mask parameter to a structure built as in the example files, for instance `p = cllc_dcx_param_example(param.sim.tc)`. The Solver Configuration of each side uses the local solver with Backward Euler and the step `p.pwm.tc`. For power from side 2 to side 1, `p.ctrl.sr_mode_1 = 1` and `p.ctrl.sr_mode_2 = 0` swap the roles of the bridges; the soft start is enabled by giving both `p.pwm.f_start` and `p.pwm.t_soft` (for instance $3f_{\mathrm{pwm}}$ down to f_{pwm} in 20 ms), and is the normal way to start a DCX from discharged DC links, where the first pulses would otherwise charge the output through the tank at the resonant peak.

In the SST bench of [1] two instances are used, one per module, with the parameters `param.cllc.p_mod1` and `param.cllc.p_mod2` built by `private/sst_cllc_ttype_setup.m` from the numbers of `private/sst_cllc_ttype_param.m`; the DC-link capacitors of side 2 are left to the inverter blocks, the two DCX can be interleaved by a quarter of the switching period (`param.cllc.interleave`, carrier phase $\pi/2$ on module 2 through `p.pwm.phase`, off by default) so that their $2f_{\mathrm{s}}$ ripple currents cancel on the shared battery bus, and both run at the sample time of the model.

5.6. Differences from the bench model

The block reproduces the bench of Section 4 with three differences. The switches are numerical (no switching losses, no thermal model), so the losses of a run are the conduction losses on R_{on}, the tank resistances, R_{fe} and the numerical damping of Section 4.4, which applies to the block in the same way; the loss budget of Table 3 is checked with the bench, not with the block. The measurements are in pu and the DC quantities are acquired as period averages, as a digital controller would; the bench logs the physical signals. The optional DC links and chokes allow the block to be used with external DC-link components (as in the SST bench, where side 2 is the DC link of the inverter block), and the two sides keep one Electrical Reference each, so that the isolation of the two DC links is preserved in the model as in the hardware.

6. Open points

- **Runs.** No run is saved: the checks of Table 7 are the first work to do on the benches, at $t_c = 50\,\text{ns}$ with C_{oss} on, and the results have to be added to this note.
- **Transformer.** The Steinmetz exponents and the AC resistance factor used by the sizing script are assumptions anchored to one datasheet point; the leakage of $4\,\mu\text{H}$ and the gap for L_m have to be confirmed with the chosen core material and winding arrangement, and the discrete part of L_r has to be designed (about $10\,\mu\text{H}$ per side at 71 A rms and 50 kHz).
- **Tolerances.** $f_r \pm 7\,\%$ moves f_n between 0.89 and 1.02: above resonance the rectifier loses the soft turn-off. If the tolerances of the tank cannot be held, f_n should be lowered to about 0.92 at the cost of a higher peak current, or the switching frequency trimmed on the measured resonance.
- **DC link of side 1.** In the module the 800 V link is shared with the inverter and sized on the 100 Hz ripple; the 100 kHz ripple current of the DCX (34 A rms) must be covered by the film part of that bank.
- **Start-up and protection.** The soft start of the library block (frequency sweep) is to be verified on the bench; the behaviour of the open-loop stage on a short circuit of side 2 (the tank current limited by Z_0 only, with $u_1/Z_0 \simeq 200\,\text{A}$ of peak) needs a protection strategy, which the inverter stage cannot provide.
- **Semiconductor option.** The $6.6\,\text{m}\Omega$ module has margin; the $13\,\text{m}\Omega$ to $16\,\text{m}\Omega$ option costs 70 W to 100 W per bridge and is to be evaluated against the module price.
- **Bench file state.** The frequency switch left in the parameter file of the 800 V bench (Section 4.4) should be resolved: either the 50 kHz line is restored or the folder is renamed to what it runs.

7. Conclusions

The isolated stage of the SST module is a CLLC resonant converter run as a DC transformer: 800 V to 800 V, 50 kW, 50 kHz with the series resonance at 52.6 kHz, a symmetrical tank of $12\,\mu\text{H}$ and $0.762\,\mu\text{F}$ per side, a $1:1$ transformer on three stacked UU93/152/30 sets with $400\,\mu\text{H}$ of magnetizing inductance for the ZVS of the driving bridge within the 300 ns dead time, and the same 1200 V SiC module on both bridges. The rectifying bridge follows the sign of its own current, the stage runs in open loop and the estimated losses are 365 W, 99.3 % of efficiency. The reason why the stage is not regulated is quantitative: near resonance the tank reactance is $-0.81\,\Omega$, the nominal power corresponds to a phase displacement of 4.5°, less than the dead time, and a regulated phase would force the rec-



tifier to switch with current. The 35 kHz variant 790 V to 700 V, born from a reading of the model at a coarse solver step, is kept as a second design case at the same power on a heavier transformer. Both stages have a bench in `solid-state-transformers`, built with the discrete blocks of the library, and a parameter set for the compact block `cllc_dcx` of `pwr-control/library`, which packs the whole stage into one Simscape component and one MATLAB Function for the model of the complete SST. The numerical damping of the fixed-step Backward Euler solver is the one modelling point to keep in mind: 2.5 Ω per microsecond of step in series with a tank whose physical resistance is 43 mΩ.





A. Files

Table 12: Files of the two repositories referred to in this note.

File	Content
`solid-state-transformers`, folder `theory_analysis_solid_state_transformer/sst_cllc_dcx_800V_800V_50kW` (the variant in `sst_cllc_dcx_790V_700V_50kW`)	
`README.md`, `cllc_dcx_sizing.md`	description of the bench and sizing note in short form
`sst_cllc_dcx.slx` (`sst_cllc_dcx_35kHz.slx`)	Simulink/Simscape model of the bench
`init_model.m`	simulation settings (`param.sim`: `tc`, `sink`, `use_coss`, run length), builds `param`
`private/cllc_dcx_param.m`	numbers of the DCX: operating point, tank, transformer, DC links, SR modes, sink modes
`cllc_analysis.m`	FHA gain, input impedance at 10 / 50 / 100 % load, $P(\varphi)$
`transformer_sizing.m`	transformer: core stack, turns, gap, litz, losses, leakage
`power_loss_calculus.m`	checks a run against the sizing: powers, currents, ZVS, losses, numerical damping
`plotting_results.m`, `spectrum.m`	figures of a run, spectra
`non_virtual_blocks_counting.m`	counts the nonvirtual blocks of the model
`library`, folder `library/masked_models/cllc_dcx/`	
`cllc_dcx.slx`	the block `cllc_dcx_mod1`
`private/cllc_dcx_ctrl_pwm.m`	source of the MATLAB Function (modulator, SR logic, control)
`update_cllc_dcx.m`	builds and refreshes the block from the sources
`cllc_dcx_param_example.m`	parameter structure of the 50 kHz stage
`cllc_dcx_param_example_35kHz.m`	parameter structure of the 35 kHz variant
`cllc_dcx_log_index.m`	indices of `ctrl_log`
`library`, folder `library/foundation/pwr_electronics/`	
`+cllc_ssc/cllc_dcx_single_phase.ssc`	Simscape component of the power stage
`+cllc_ssc/onoff.m`	enumeration of the options of the component
`cllc_ssc_lib.slx`	compiled library of the component (`ssc_build cllc_ssc`)
`library`, folder `library/documentation/cllc_dcx/`	
`cllc_dcx.tex`, `cllc_dcx.pdf`	this note, source and PDF (the same source as in `repositories-documentation`)
`figures/*.dat`, `py/cllc_dcx_figures.py`	data files of the analytic figures and the script that produces them



The data files of Figures 2 and 4 (`figures/fha_gain_50kHz.dat`, `figures/tank_waveforms_50kHz.dat`, and the same two for the variant) are produced by `py/cllc_dcx_figures.py` from the numbers of Table 4; the script also prints the derived quantities of both parameter sets (resonance, tank, FHA gain and input impedance at three loads, $X(f_s)$, R_{num} per unit of step).

References

[1] D. Bagnara, *Solid State Transformers: architecture, modelization and control*, pwr-control technical report, Rev. 01, October 2026, `repos_doc/solid_state_transformers` of https://github.com/pwr-control/repositories-documentation.

[2] D. Bagnara, *Single-Phase Grid-Tied T-Type Inverter: model, control design and discrete implementation*, pwr-control technical report, October 2026, `repos_doc/single_phase_grid_tied_inverter_analysis` of https://github.com/pwr-control/repositories-documentation.

[3] pwr-control, *solid-state-transformers* repository, https://github.com/pwr-control/solid-state-transformers.

[4] pwr-control, *library* repository (Simscape components, masked blocks and documentation), https://github.com/pwr-control/library.

[5] J.-H. Jung, H.-S. Kim, M.-H. Ryu, J.-W. Baek, "Design methodology of bidirectional CLLC resonant converter for high-frequency isolation of DC distribution systems," *IEEE Transactions on Power Electronics*, vol. 28, no. 4, pp. 1741–1755, April 2013.

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[7] B. Yang, F. C. Lee, A. J. Zhang, G. Huang, "LLC resonant converter for front end DC/DC conversion," *Proc. IEEE APEC*, 2002, pp. 1108–1112.

[8] H. Huang, "Designing an LLC resonant half-bridge power converter," Texas Instruments Power Supply Design Seminar SEM1900, Topic 3, 2010.

[9] R. W. De Doncker, D. M. Divan, M. H. Kheraluwala, "A three-phase soft-switched high-power-density DC/DC converter for high-power applications," *IEEE Transactions on Industry Applications*, vol. 27, no. 1, pp. 63–73, 1991.

[10] F. Krismer, *Modeling and Optimization of Bidirectional Dual Active Bridge DC–DC Converter Topologies*, PhD thesis, ETH Zürich, 2010.



[11] MathWorks, “Making optimal solver choices for physical simulation” and “Solver Configuration,” Simscape documentation.

[12] Wolfspeed, *CAB006M12GM3 1200 V, 6.6 mΩ silicon carbide half-bridge module*, datasheet.